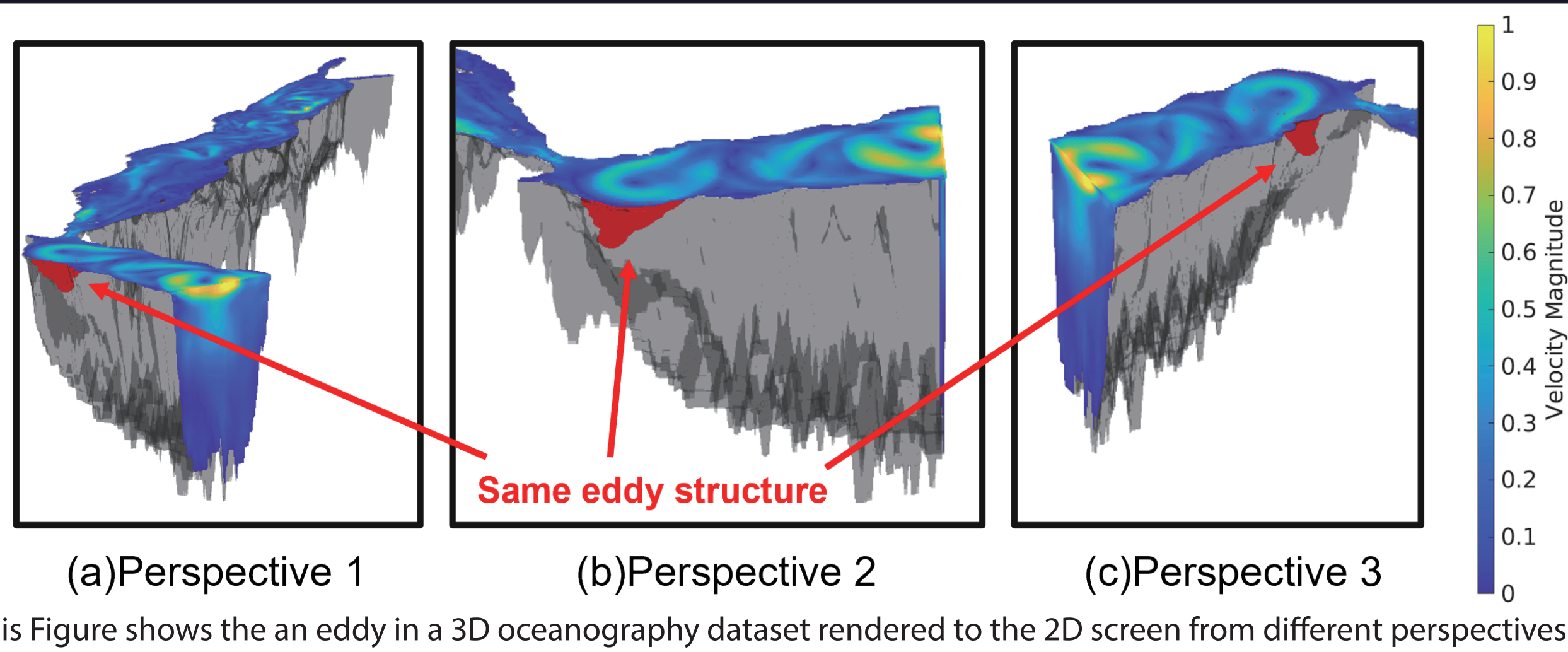


Using 2D deep neural networks to segment and visualize 3D features in scientific simulations

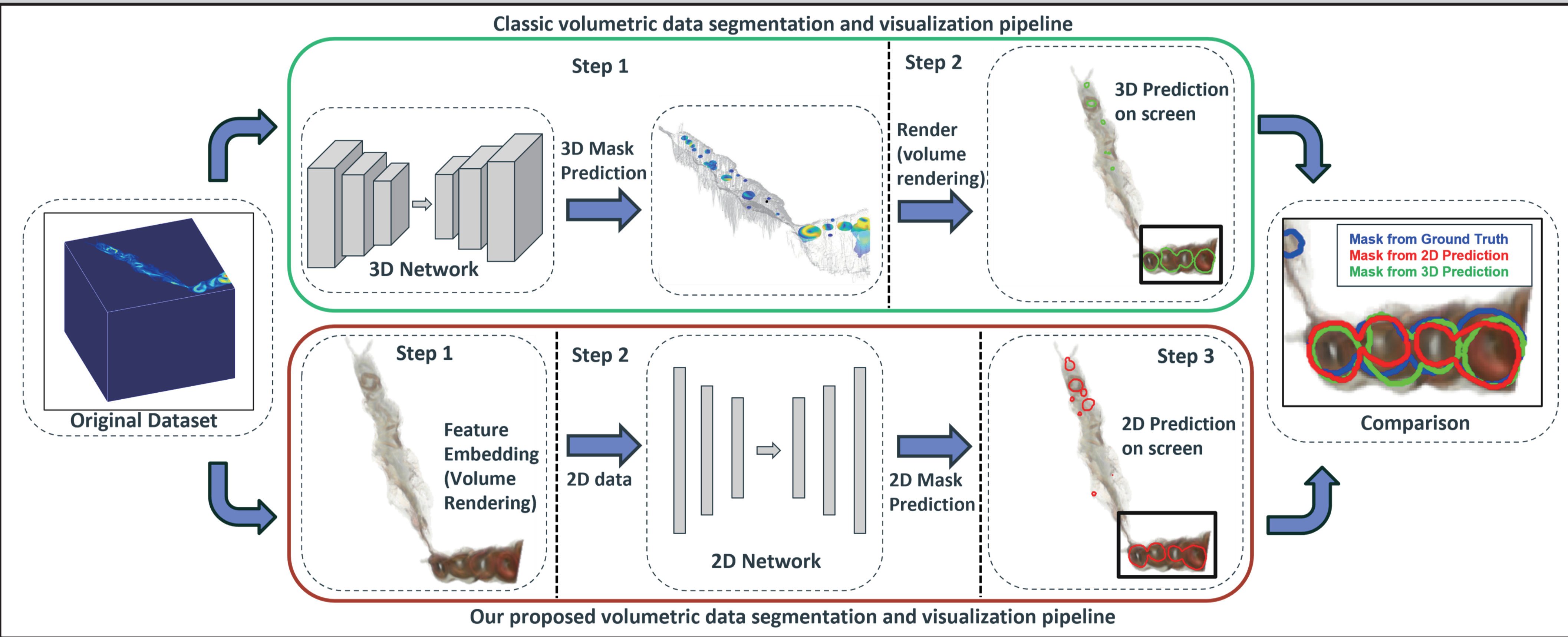
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Research Question

- 3D segmentation with neural networks is costly — e.g., training on a 857×857×35×36 time-varying dataset can take ~20 days. How can we speed this up?
- Since the final output is ultimately displayed as 2D renderings, performing segmentation after rendering emerges as a practical and efficient alternative.

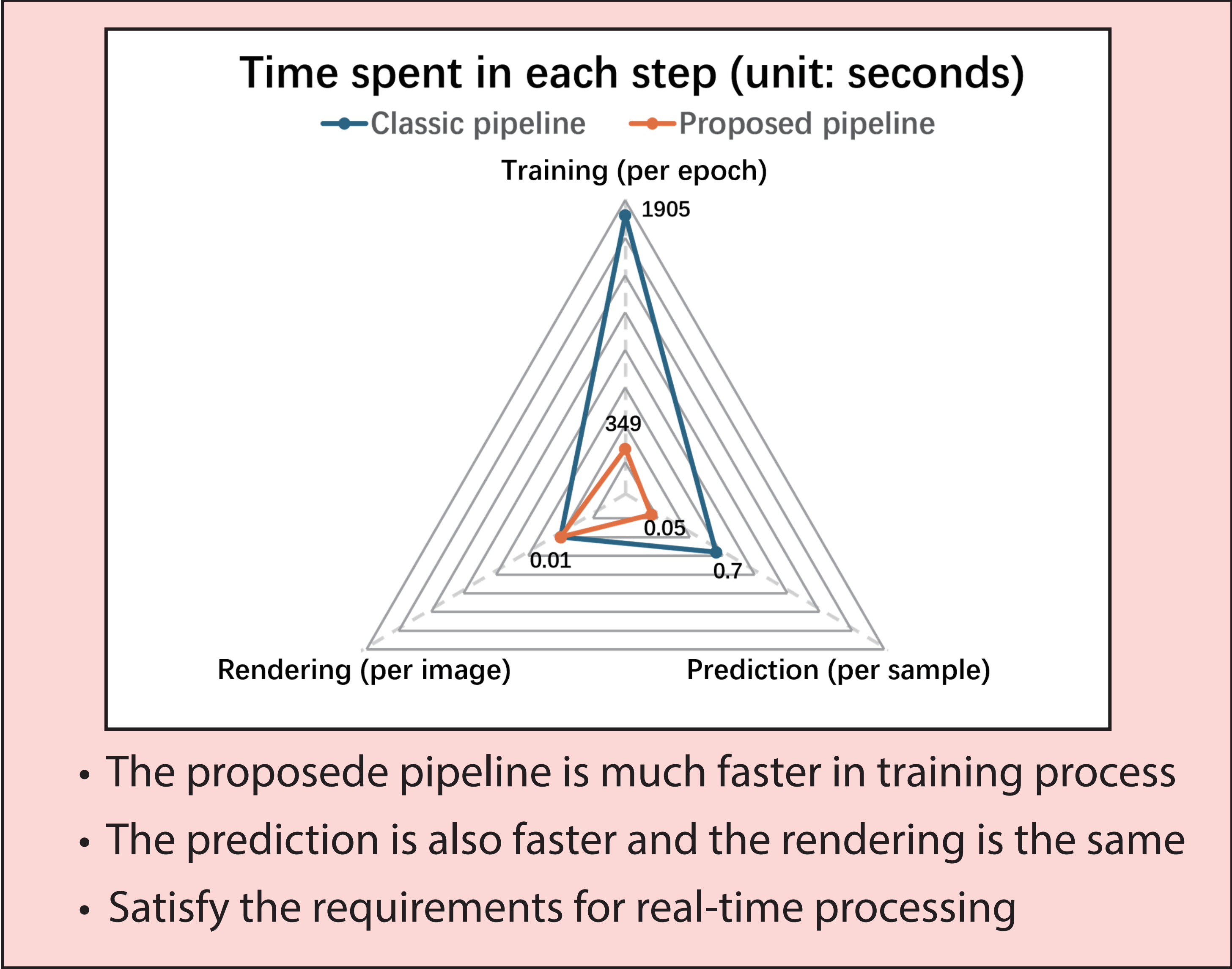
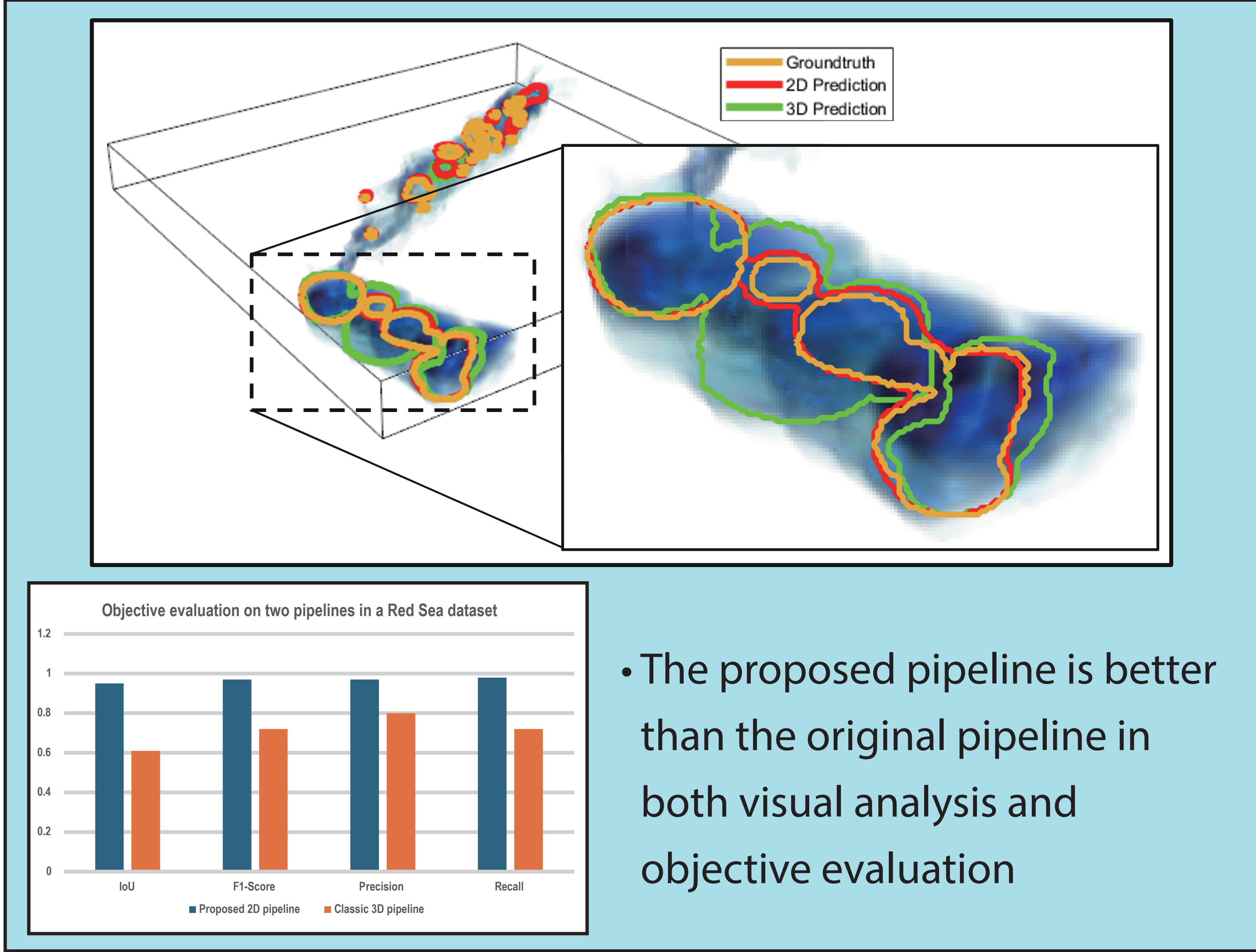


Methodology



- The traditional approach in the classic pipeline uses a 3D neural network to segment the original dataset, and the segmentation output is rendered for visualization
- As a alternative approach in the proposed pipeline, the data is rendered into a 2D image first, then a 2D network is applied to perform segmentation directly on the rendered view

Main findings



Limitations and future work

- The impact of the transfer function has been investigated, but remains not fully understood
- A comprehensive view requires predictions from multiple perspectives
- While the pipeline yields markedly superior results on our dataset, its comparable performance on other datasets remains effective and consistent with expectations.
- Incorporating 3D information may further enhance performance.

